

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.2s gimbal=0.0235279005s cost=861.25480425 lag_gradient={'rotor':
-2.4119008101325425, 'gimbal': -4.732455643252642e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.3877787807846427e-17,
'analytic_l2_norm': 8.142562980940385, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 8.142562980940385, 'relative_error': 1.0}, {'direction':
'rotor_delay_seconds', 'scheme': 'forward', 'step': 1.3877787807814457e-17,
'analytic_l2_norm': 8.14257103917186, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 8.14257103917186, 'relative_error': 1.0}, {'direction':
'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 18.497290807553128, 'finite_difference_l2_norm': 18.497290632331477,
'difference_l2_norm': 8.063596137256603e-07, 'relative_error': 4.35933900869631e-08}]
smooth width=2: rotor=0.2s gimbal=0.0216392035s cost=861.257987026 lag_gradient={'rotor':
-7.59084995902804, 'gimbal': 3.4453170294490576e-07} FD=[{'direction': 'negative_gradient',
'scheme': 'forward', 'step': 8.187894806675134e-16, 'analytic_l2_norm': 8.968256086104226,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 8.968256086104226, 'relative_error':
1.0}, {'direction': 'rotor_delay_seconds', 'scheme': 'forward', 'step':
8.187894806610529e-16, 'analytic_l2_norm': 8.968262533024147, 'finite_difference_l2_norm':
0.0, 'difference_l2_norm': 8.968262533024147, 'relative_error': 1.0}, {'direction':
'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 20.461580641471496, 'finite_difference_l2_norm': 20.46157834612638,
'difference_l2_norm': 4.888453914326914e-06, 'relative_error': 2.389089093351373e-07}]
smooth width=1: rotor=0.2s gimbal=0.0199908414s cost=861.258952124 lag_gradient={'rotor':
-8.000840801640093, 'gimbal': -1.0418750194624002} FD=[{'direction': 'negative_gradient',
'scheme': 'forward', 'step': 3.4635687355268646e-11, 'analytic_l2_norm': 11.65467515329636,
'finite_difference_l2_norm': 11.65392177117251, 'difference_l2_norm': 0.15651207349007307,
'relative_error': 0.013429123629053345}, {'direction': 'rotor_delay_seconds', 'scheme':
'forward', 'step': 3.4342542698517775e-11, 'analytic_l2_norm': 11.15297267251353,
'finite_difference_l2_norm': 11.153469524409548, 'difference_l2_norm':
0.0005195367038920389, 'relative_error': 4.658072564371332e-05}, {'direction':
'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 26.073079846355956, 'finite_difference_l2_norm': 26.07482386037388,
'difference_l2_norm': 0.608045010991585, 'relative_error': 0.023319237523810696}]
smooth width=0.5: rotor=0.19913685s gimbal=0.0193988957s cost=861.259893521
lag_gradient={'rotor': -2.5825566285266683, 'gimbal': 0.5153157573847573} FD=[{'direction':
'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 17.625718927453033, 'finite_difference_l2_norm': 17.622890582037858,
'difference_l2_norm': 0.34083652532692743, 'relative_error': 0.01933745379293753},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 15.08811320804464, 'finite_difference_l2_norm': 15.088068409399652,
'difference_l2_norm': 8.428259238888978e-05, 'relative_error': 5.586025981297132e-06},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 38.78063625368655, 'finite_difference_l2_norm': 38.63061636791167,
'difference_l2_norm': 1.8055391107017817, 'relative_error': 0.046557748534364084}]
strict profile iterations: 1
iteration 1: best=(0.199136850355, 0.019398895719) cost=861.25474062 rotor_range=(0.1941391201,
0.2) gimbal_range=(0.01440021179, 0.024397579648) expansions=[]

selected rotor lag 0.199136850355 s (39.8455 publish periods)
selected gimbal lag 0.019398895719 s (3.8808 publish periods)
selected objective cost 861.254738705